

1 REQUIRED INPUTS

Wall geometry	wall center coordinates (x_i, y_i) , length L_i , height h_i , thickness t_i
Wall weight	wall or tributary story weight W_i for each wall segment
Material	elastic modulus E_i for each wall segment
Loads	base shears $F_{P,x}$, $F_{P,y}$ and torsions T_x^+ , T_x^- , T_y^+ , T_y^-
Order: CM $\rightarrow$ relative rigidity $\rightarrow$ CR $\rightarrow$ torsion property $\rightarrow$ wall forces.	

2 S1 CENTER OF MASS

Compute wall weight takeoff first; then sum weight moments.

$$xW_i = x_iW_i, \quad yW_i = y_iW_i \quad (1)$$

i	wall segment index
xW_i	x -coordinate weight moment for wall segment i
x_i	x coordinate of wall segment i
W_i	wall or tributary story weight for wall segment i
yW_i	y -coordinate weight moment for wall segment i
y_i	y coordinate of wall segment i

$$x_1 = \frac{\sum_i xW_i}{\sum_i W_i}, \quad y_1 = \frac{\sum_i yW_i}{\sum_i W_i} \quad (2)$$

x_1	center of mass x coordinate
y_1	center of mass y coordinate

3 S2 RELATIVE RIGIDITY

Use the same trial lateral force for each wall.

$$\Delta_i = \frac{4P(h_i/L_i)^3}{E_i t_i} + \frac{3P(h_i/L_i)}{E_i t_i}, \quad P = 1,000,000 \text{ lb} \quad (3)$$

Δ_i	lateral deflection of wall segment i under trial force P
P	trial lateral force used for relative rigidity
h_i	wall height for wall segment i
L_i	wall length for wall segment i
E_i	elastic modulus for wall segment i
t_i	wall thickness for wall segment i

$$R_i = \frac{1}{\Delta_i}, \quad R_{x,i} = R_i \ (i \in X), \quad R_{y,i} = R_i \ (i \in Y) \quad (4)$$

R_i	relative rigidity of wall segment i
$R_{x,i}$	relative rigidity for an X -wall segment
$R_{y,i}$	relative rigidity for a Y -wall segment
X	wall set used with R_x and E-W force distribution
Y	wall set used with R_y and N-S force distribution

$$\Sigma R_x = \sum_{i \in X} R_{x,i}, \quad \Sigma R_y = \sum_{i \in Y} R_{y,i} \quad (5)$$

ΣR_x	total relative rigidity in the x direction
ΣR_y	total relative rigidity in the y direction

4 S3 CENTER OF RIGIDITY

$$xR_{y,i} = x_iR_{y,i}, \quad yR_{x,i} = y_iR_{x,i} \quad (6)$$

$xR_{y,i}$	x -coordinate rigidity moment for Y -wall segment i
$yR_{x,i}$	y -coordinate rigidity moment for X -wall segment i

$$x_r = \frac{\sum_{i \in Y} x R_{y,i}}{\sum R_y}, \quad y_r = \frac{\sum_{i \in X} y R_{x,i}}{\sum R_x} \quad (7)$$

x_r center of rigidity x coordinate
 y_r center of rigidity y coordinate

5 S4 TORSIONAL PROPERTIES

$$d_{x,i} = x_i - x_r, \quad d_{y,i} = y_i - y_r \quad (8)$$

$d_{x,i}$ x distance from wall segment i to center of rigidity
 $d_{y,i}$ y distance from wall segment i to center of rigidity

$$RD_{y,i} = R_{y,i} d_{x,i}, \quad RD_{x,i} = R_{x,i} d_{y,i} \quad (9)$$

$RD_{y,i}$ torsional force coefficient for Y -wall segment i
 $RD_{x,i}$ torsional force coefficient for X -wall segment i

$$J_R = \sum_{i \in Y} R_{y,i} d_{x,i}^2 + \sum_{i \in X} R_{x,i} d_{y,i}^2 \quad (10)$$

J_R total torsional rigidity term, equal to $\Sigma(RD^2)$

6 S5A N-S EQ FORCES

For N-S force distribution, use Y -walls and R_y .

$$F_{v,y,i} = \frac{R_{y,i}}{\sum R_y} F_{P,y} \quad (11)$$

$F_{v,y,i}$ direct shear force in Y -wall segment i
 $F_{P,y}$ N-S base shear

$$F_{t,y,i}^+ = \frac{RD_{y,i}}{J_R} T_y^+, \quad F_{t,y,i}^- = \frac{RD_{y,i}}{J_R} T_y^- \quad (12)$$

$F_{t,y,i}^+$ torsional force in Y -wall segment i from T_y^+
 T_y^+ positive N-S torsion
 $F_{t,y,i}^-$ torsional force in Y -wall segment i from T_y^-
 T_y^- negative N-S torsion

$$F_{1,y,i} = F_{v,y,i} + F_{t,y,i}^+, \quad F_{2,y,i} = F_{v,y,i} + F_{t,y,i}^- \quad (13)$$

$F_{1,y,i}$ total N-S force in Y -wall segment i using T_y^+
 $F_{2,y,i}$ total N-S force in Y -wall segment i using T_y^-

$$F_{\text{design},y,i} = \max(F_{1,y,i}, F_{2,y,i}) \quad (14)$$

$F_{\text{design},y,i}$ N-S design force in Y -wall segment i

7 S5B E-W EQ FORCES

Mirror S5A for E-W force distribution: use X -walls and R_x .

$$F_{v,x,i} = \frac{R_{x,i}}{\sum R_x} F_{P,x} \quad (15)$$

$F_{v,x,i}$ direct shear force in X -wall segment i
 $F_{P,x}$ E-W base shear

$$F_{t,x,i}^+ = \frac{RD_{x,i}}{J_R} T_x^+, \quad F_{t,x,i}^- = \frac{RD_{x,i}}{J_R} T_x^- \quad (16)$$

$F_{t,x,i}^+$ torsional force in X -wall segment i from T_x^+
 T_x^+ positive E-W torsion
 $F_{t,x,i}^-$ torsional force in X -wall segment i from T_x^-
 T_x^- negative E-W torsion

$$F_{1,x,i} = F_{v,x,i} + F_{t,x,i}^+, \quad F_{2,x,i} = F_{v,x,i} + F_{t,x,i}^- \quad (17)$$

$F_{1,x,i}$ total E-W force in X -wall segment i using T_x^+
 $F_{2,x,i}$ total E-W force in X -wall segment i using T_x^-

$$F_{\text{design},x,i} = \max(F_{1,x,i}, F_{2,x,i}) \quad (18)$$

$F_{\text{design},x,i}$ E-W design force in X -wall segment i

CHECK ORDER

- 1 Compute W_i , xW_i , yW_i , then x_1 and y_1 .
- 2 Compute Δ_i and R_i for each wall; split into $R_{x,i}$ and $R_{y,i}$ wall sets.
- 3 Compute x_r from Y -walls and y_r from X -walls.
- 4 Compute $d_{x,i}$, $d_{y,i}$, $RD_{x,i}$, $RD_{y,i}$, and $J_R = \Sigma(RD^2)$.
- 5 Distribute $F_{P,y}$ to Y -walls with T_y^+ and T_y^- .
- 6 Mirror for $F_{P,x}$ to X -walls with T_x^+ and T_x^- .